Relative safety of first-time volunteer and replacement donors in West Africa

Jean-Pierre Allain, Francis Sarkodie, Kwame Asenso-Mensah, and Shirley Owusu-Ofori

BACKGROUND: A significantly higher level of safety between nonremunerated volunteer and replacement donor blood is assumed. This is supported by global data without stratifying between genuine replacement and paid donors, for first-time or repeat volunteer, or according to age.

STUDY DESIGN AND METHODS: In 2008, first-time volunteer and replacement donors were identified, and confirmed human immunodeficiency virus antibody (anti-HIV), hepatitis B surface antigen (HBsAg), and hepatitis C virus antibody (anti-HCV)-positive screening results were collated. Data were analyzed according to age and sex between the two types of donors.

RESULTS: In 6640 first-time volunteer and 4360 replacement donors, the prevalence of anti-HIV and HBsAg (1.03 and 13.8% vs. 1.1 and 14.9%, respectively) was not significantly different. Anti-HIV prevalence was higher in replacement donors less than age 20 than in first-time volunteers; the difference was not significant. HBsAg and anti-HIV confirmed-positive prevalence was significantly higher in first-time volunteer donors over age 20.

CONCLUSION: In Kumasi, Ghana, viral safety of replacement and first-time volunteer donors was similar, constituting a single population of donors. Safety increment is provided by repeat donation applicable to either group, through different approaches. A blood unit from replacement donor costs half or less than that from a volunteer donor; similar studies conducted elsewhere in sub-Saharan Africa may lead to changes in current strategies.

For many years, WHO and many western transfusion organizations have advocated the necessity of recruiting exclusively volunteer donors and eliminating replacement donors (individuals prompted to provide blood unit(s) in replacement of the blood transfused to one of their relatives or friends) stating that the former were safer than the latter.[1] In the past, replacement donors were often a mix of paid and genuine family replacement donors. Paid donors are widely recognized to be "unsafe" particularly for HIV and this contamination of replacement donors did make the group unsafe. However, in most cases, upon such evidence, hospitals and blood services made substantial efforts to eliminate paid donors leaving genuine replacement donors as the main source of blood in sub-Saharan Africa.[2,3] There is therefore a need to reexamine the evidence and policy around the safety characteristics of replacement donors.

Despite the largely and forcefully advocated concerns regarding replacement donor blood safety, there are two main reasons why replacement donation remains dominant in the African continent. One is that family and community ties are considerably stronger than in other types of societies, making the gift of blood a natural contribution to relieve sufferers in hospitals.[4] It is also the least expensive source of blood by a wide margin ($12-$18/unit). Recent data have demonstrated that costs for recruitment of donors, organization and implementation of mobile blood collection sessions, advertisement for volunteer blood donors, and infrastructure for rewarding volunteer donation at the donation site and more publicly for repeat donors make the cost of volunteer blood at least

From the Division of Transfusion Medicine, Department of Haematology, University of Cambridge, Cambridge, UK; and the Transfusion Medicine Unit, Komfo Anokye Teaching Hospital, Kumasi, Ghana.

Address reprint requests to: Prof. Jean-Pierre Allain, Department of Haematology, Cambridge Blood Centre, Long Road, Cambridge CB2 2PT, UK; e-mail: jpa1000@cam.ac.uk.

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twice more expensive than replacement blood ($36-$52).[2] When this approach is associated with centralization, automation, large operations, and following the standard model of Western transfusion, the costs then triple or quadruple (>$80/unit), becoming unaffordable for most sub-Saharan countries where income per capita is below $1000/annum.[3]

The objective of this study was to compare the epidemiology of viral infections in first-time blood donors and replacement donors when adjusted according to age. The underlying assumption was that the reported safety difference (when paid donors were excluded) was mostly related to age so that age adjustment would provide a more accurate estimation of the assumed additional risk.

MATERIALS AND METHODS

In 2008 at the Komfo Anokye Teaching Hospital (KATH) in Kumasi, Ghana, replacement and first-time volunteer donors were identified from written records (since 2003, the small number of paid donors was expelled from the hospital ground by the security staff). Age and sex were recorded. For each type of donor, rapid tests for hepatitis B surface antigen (HBsAg; Determine, Abbott Laboratories, Delkenheim, Germany), human immunodeficiency virus antibody (anti-HIV, Determine anti-HIV, Abbott), and hepatitis C virus antibody (anti-HCV; Intec, Hong Kong) were performed before donation as previously described.[5] Briefly, after answering health questions and satisfying weight (>50 kg) and blood pressure (maximum, ≤140 mmHg) requirements, 2 to 3 mL of venous whole blood from all types of donors was collected, clotted, and centrifuged and serum was tested with three rapid tests for HBsAg, anti-HIV, and anti-HCV. Nonreactive donors were suitable for blood donation. Donors with reactivity to any of these assays were excluded from donation and advised to attend the hospital blood center to collect the test results after the confirmatory procedure was completed, as previously described.[6] Anti-HIV and anti-HCV were confirmed with enzyme immunoassays (EIAs) in microtiter plates (Bio-Rad, Marnes la Coquette, France). Samples with concordant reactivity of the rapid test and the EIA were considered confirmed.

The significance of viral marker distributions between the two groups was determined by two-sided Mann-Whitney test for nonparametric data. Differences in seroprevalence between groups were tested for significance by means of 3×2 chi-square test.

RESULTS

Results are presented in Table 1 and Fig. 1. Confirmation was performed on all reactive samples on the leftover serum from the initial tested blood. In 2008, 60% of volunteer and 40% of replacement donors entered the previously described donor care program.[6] The differences in age and sex distribution between first-time volunteers and replacement donors were as previously described (replacement donors—median age 31 years, 12.6% female; volunteer donors—median age 19, 38.8%

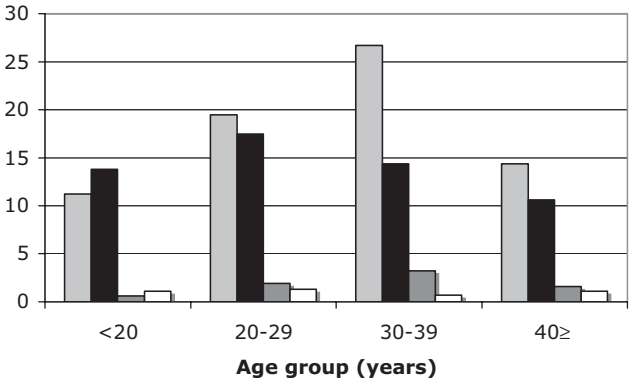


Fig. 1. Age distribution of HBsAg and anti-HIV prevalence in first-time volunteer and replacement Ghanaian blood donors. HBsAg prevalence in first-time volunteer donors (□) and in replacement donors (■) are shown. Prevalence of anti-HIV confirmed positive in first-time volunteer donors (▩) and in replacement donors (□).

TABLE 1. HBV and HIV prevalence in first-time volunteer and replacement Ghanaian blood donors according to age

	Age group (years)				
Parameters studied	<20	20-29	30-39	40≥	Total (%)
Number of first-time volunteers	4685 (70.6)	1423 (21.4)	345 (5.2)	187 (2.8)	6640
Male/female	2675/2010	988/435	274/71	130/57	4067/2573 (61.3)
HBsAg positive (%)	523 (11.2)	277 (19.5)	92 (26.7)	27 (14.4)	919 (13.8)
Anti-HIV confirmed*	28 (0.6)	27 (1.9)	11 (3.2)	3 (1.6)	69 (1.03)
Anti-HCV confirmed	1	6	4	0	11 (0.17)
Number of replacement volunteers	274	1879	1440	767	4360
Male/female	244/30	1623/256	1291/149	676/91	3834/526 (87.9)
HBsAg positive (%)	31 (11.3)	329 (17.5)	208 (14.4)	81 (10.6)	649 (14.9)
Anti-HIV confirmed	3 (1.1)	25 (1.3)	10 (0.7)	11 (1.4)	50 (1.1)
Anti-HCV confirmed	1	6	4	2	13 (0.36)

* Confirmed = confirmed by concordant reactivity of two different screening assays.

female).[7] As described in the same donor population in 2003,[8] first-time volunteer and replacement donors had a similar prevalence of HBsAg (13.8 and 14.9%, respectively), consistent with hepatitis B virus (HBV) being transmitted vertically or during infancy.[9] Surprisingly, the prevalence of HBsAg carriage in first-time volunteer donors over age 20 was significantly higher than in replacement donors by a wide margin (20.3 and 15.1%, respectively; $p < 0.0001$). The overall prevalence of confirmed anti-HIV was not significantly higher in the replacement donor group than in the first-time volunteer group (1.1% vs. 1.03%, respectively). When stratified by age groups, the prevalence of anti-HIV in first-time volunteer donors below 20 years of age was not significantly lower than in replacement donors ($p = 0.31$) while above 20 years of age it was higher than in replacement donors (2.1% vs. 1.1%, respectively; $p < 0.05$).

DISCUSSION

Data published in the literature, including our own, reported a prevalence of anti-HIV, HBsAg, and anti-HCV significantly lower in volunteer donors than replacement donors.[7] However, in most studies, differentiation between first-time and repeat volunteer donors was not made. Second, most volunteer donors in sub-Saharan Africa are secondary school students with a median age below 20 while median age of replacement donors was 30 to 32 years.[7,8] It is recognized that the risk of HIV infection in sub-Saharan Africa is highest for men 30 to 40 and for younger women of 20 to 29.[10,11] Females constitute a minority of approximately 10% in replacement donors and a higher proportion in volunteer donors who are also recruited in female schools constituting 30% to 40% of volunteer donors.[12] These various biases were generally not emphasized in published studies. There is clear evidence that repeat donation brings a significant increase in blood safety and this could be equally applied to volunteer and to replacement donors.[7] The experience of most blood centers in sub-Saharan Africa is that the rate of repeat donation in volunteer donors in schools remains low (approx. 25% in Kumasi, Ghana). The majority of donors are therefore either first-time volunteer or replacement donors.[11]

The data collected in Kumasi, Ghana, in 2008 revealed no significant difference in prevalence of viral markers between first-time volunteer and replacement donors. However, when stratified according to age, differences became apparent. While volunteer first-time donors below age 20 had lower prevalence of both HBsAg and confirmed anti-HIV than replacement donors, the differences were not significant. In contrast, both HBsAg and anti-HIV prevalence was significantly higher in older (>20) first-time volunteer donors than in replacement donors. The HIV data obtained in Ghana are similar to what was previously published in Brazil in a considerably larger number of donors.[13] Irrespective of sex or year of collection between 1995 and 2001, anti-HIV prevalence was higher in first-time volunteer than replacement donors. In Kumasi, the incremental level of safety usually attributed to volunteer donors was strictly related to the younger age of the donors (70.6% below age 20; Table 1). When prevalence of both HBsAg and anti-HIV was adjusted according to age, replacement donors were safer than first-time volunteers. As a result, first-time donors, whether volunteer or replacement, provided a similar level of viral safety. Efforts made in other West African countries to eliminate replacement donors at high cost appear possibly unjustified. The current policy at Komfo Anokye Teaching Hospital blood center in Kumasi is to increase the pool of volunteer donors but keep replacement donors to improve the blood supply, particularly during school recesses when the younger student population is not available. This policy is justified by evidence. In our circumstances, both volunteer and replacement donors should be considered equally suitable first-time donors and the main justification for continuing to increase the percentage of volunteer donors becomes that it is easier to convert them into repeat donors. In this study and for the past 10 years, no effort was put into attempting to attract replacement donors to repeat donation. As a result while overall 24% first-time volunteer donors repeated donation in 2008, less than 1% replacement give blood a second time spontaneously. In some ways, devising a strategy to convert replacement donors into repeating donation would add to costs and be of uncertain efficacy, therefore diminishing the attractive cost differential with volunteer blood but would certainly increase viral safety. In the Kumasi setting, the standard cost per unit of blood excluding triplex nucleic acid test (NAT) was $13.4 and $17.4/unit including NAT. The added cost per volunteer unit was approximately $12.5/unit. To this total cost of $26 to $30/unit of voluntary blood, $1.5 was added when collecting blood in the previously described setting of an FM radio station.[12]

The blood centers' policy of excluding all replacement donors practiced in resource-poor settings incurs high costs and risks of reducing the already critically short blood supply (<10 units/1000 inhabitants in most sub-Saharan countries). It is therefore important to repeat this study in other resource-poor settings where paid donors have been eliminated to prove or disprove the evidence obtained in Ghana.[11,14] If our findings are replicated, the donor selection policy in the future may focus on strategies to retain both volunteer and replacement donors as repeat donors, which would significantly improve the safety of the donor pool and improve the blood supply.[12] We do not know whether it would be more difficult to incite replacement donors to repeat donation than the volunteers and, to our knowledge, there is no program in

sub-Saharan Africa that has attempted to address this issue. However, our experience of partnership with FM radio stations, where providing an attractive environment to blood donation spontaneously led to a 65% rate of repeat donations in a population of donors of similar age and sex to replacement donors, suggests that such task might not be impossible if innovatively considered for replacement donors.

CONFLICT OF INTEREST

None of the authors declare any conflict of interest regarding this manuscript.

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